

b) Stack Allocation:-

- Memory is allocated during function calls.
- Uses stack (LIFO structure).

Characteristics:-

- Automatically allocated and deallocated.
- Each function gets an activation record.

Used in:-

- Local variables.
- Function calls.
- Recursion.

Advantages:-

- Efficient memory usage.
- Automatic management.

Disadvantages:-

- Limited size.
- Cannot store dynamic data.

Eg:- void fun () {
 int x = 5;
}

c) Heap Allocation:-

- Memory is allocated dynamically at runtime.

Characteristics:-

- Managed manually or by garbage collector.
- Flexible size.